

mrm-mdd-kit — Project Brief

What This Is

`mrm-mdd-kit` is an open source GitHub repository that provides a reusable Model Development Document (MDD) kit for developers building ML and LLM applications in regulated industries — specifically banking, insurance, and finance.

The kit is designed to be consumed as a **git submodule** inside any ML/LLM project repo. It gives developers everything they need to produce a compliant MDD that satisfies **SR 11-7** (Federal Reserve Supervisory Guidance on Model Risk Management) and its **2026 revised guidance**.

There is currently no clean, open source, SR 11-7 compliant MDD kit built for practitioners. This repo fills that gap.

Problem Being Solved

Developers building ML and LLM applications in regulated environments must go through a governance process called Model Risk Management (MRM). A core artifact of that process is the Model Development Document (MDD). Currently:

- No open source MDD template exists that is grounded in SR 11-7
 - Developers have to figure out compliance requirements from scratch
 - No one has integrated AI coding assistant support (Claude Code) into the MDD workflow
 - Teams working across multiple projects have no shared, versioned kit to stay in sync
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Audience

Primary: Developers building ML and LLM applications in regulated industries (banking, insurance, finance) who must satisfy model risk governance requirements.

Secondary: Model risk and compliance teams who want a better starting point for MDD templates.

Scope

In scope: Model Development Document (MDD) only.

Out of scope (for now): Model Validation Documents, Model Inventory templates, ongoing monitoring checklists, or any other MRM artifacts. Keep it focused.

Repo Name

mrm-mdd-kit

Full Repo Structure

```
mrm-mdd-kit/
├── README.md                ← what the kit is, who it's for, how to us
├── CHANGELOG.md             ← version history, regulatory change track
├── CONTRIBUTING.md          ← how the community contributes
├── CLAUDE.md                ← instructions for Claude Code on how to u
├── template/
│   └── MDD.md               ← blank SR 11-7 compliant MDD template
├── examples/
│   ├── llm_complaint_classifier/
│   │   ├── scenario.md      ← plain English fictional backstory
│   │   └── MDD.md           ← fully filled out fictional MDD
│   ├── fraud_detection_xgboost/
│   │   ├── scenario.md
│   │   └── MDD.md
│   ├── credit_scoring_random_forest/
│   │   ├── scenario.md
│   │   └── MDD.md
│   ├── aml_monitoring_neural_network/
│   │   ├── scenario.md
│   │   └── MDD.md
│   └── cecl_logistic_regression/
│       ├── scenario.md
│       └── MDD.md
├── governance/
│   ├── sr11-7.md            ← original SR 11-7 regulation (public Fed
│   ├── sr11-7-revised-2026.md ← 2026 revised guidance (public Fed docume
│   └── update-process.md    ← how to monitor regulatory changes and ver
└── .github/
    └── CONTRIBUTING.md      ← GitHub-specific contribution guidelines
```

Key Design Decisions

Why submodule? The kit needs to be shared across multiple project repos and kept in sync across a team. Git submodules let each project repo point to a specific versioned release of the kit. When the regulation changes and the kit is updated, teams deliberately bump their submodule — it's a conscious, traceable decision.

How submodule works in practice:

```
# Adding the kit to a new project
git submodule add https://github.com/[username]/mrm-mdd-kit _mrm-mdd-kit

# Teammates after cloning a project
git submodule update --init

# Pulling kit updates into a project
cd _mrm-mdd-kit && git pull origin main && cd ..
git add _mrm-mdd-kit && git commit -m "bump mrm-mdd-kit to latest"
```

Project CLAUDE.md imports the kit: Each project repo has its own `CLAUDE.md` that imports the kit's `CLAUDE.md`:

```
@./_mrm-mdd-kit/CLAUDE.md
```

Claude Code reads the project `CLAUDE.md` first, hits the import, and pulls in full MDD generation instructions automatically.

Generated MDD lives in the project: The `MDD.md` output file lives in the root of each project repo — not inside the kit. It gets committed and versioned alongside the project code. It's a living document that grows as the project evolves.

Why the examples are fictional — and a critical disclaimer: Banks don't publish internal compliance documents. All five example MDDs are completely made up — fictional bank names, fictional model details, fictional performance metrics, fictional personnel. This is intentional and mandatory.

 **CRITICAL — DATA PRIVACY RULE:** Never put real company data, proprietary models, internal documents, customer data, employee names, or any confidential information into this repository or any public fork of it. This kit is public and open source. Everything in it must be either publicly available regulatory content or completely fictional. When generating MDD examples, always use made-up institution names,

invented data statistics, and fictional scenarios. This rule applies to contributors, users, and Claude Code when generating content from this kit.

Vectice and other references: Vectice’s MDD guide and Eugene Yan’s ml-design-docs repo were used as structural inspiration during design. Neither is copied into this repo — content is original, derived from public regulatory sources. Both are credited in README references.

Versioning and regulatory changes: The kit uses GitHub releases with semantic version tags (v1.0.0, v1.1.0, etc.). When SR 11-7 guidance changes, the kit is updated, a new release is tagged, and CHANGELOG.md documents what changed. Downstream project submodules are notified via release notes.

The Five Examples

All examples use completely fictional institutions, people, data, and performance metrics. They exist solely to demonstrate the expected depth, tone, and structure of a completed MDD. No real company data was used.

Folder	Model Type	Use Case	Fictional Scenario
<code>llm_complaint_classifier</code>	LLM	Customer complaint classification	A mid-size regional bank uses an LLM to classify inbound customer contacts (email, chat, phone transcripts) as complaints or non-complaints, routing confirmed complaints into their CFPB workflow
<code>fraud_detection_xgboost</code>	Gradient Boosting (XGBoost)	Real-time transaction fraud scoring	A retail bank scores card transactions in real-time for fraud probability, triggering step-up authentication or automatic declines
			A consumer lending

<code>credit_scoring_random_forest</code>	Random Forest	Probability of default — personal loans	division uses a random forest to score personal loan applicants on probability of default at origination
<code>aml_monitoring_neural_network</code>	Neural Network	AML suspicious activity detection	A bank's BSA/AML team uses a neural network to flag suspicious transaction patterns for SAR filing consideration
<code>cecl_logistic_regression</code>	Logistic Regression	CECL expected credit loss forecasting	A community bank estimates lifetime expected credit losses on its consumer loan portfolio for CECL regulatory capital reporting

Build Order

Work through the kit in this sequence:

1. Create the GitHub repo, initialize with this `project-brief.md` as the only file
2. Pull SR 11-7 and 2026 revised guidance into `governance/` as markdown
3. Draft `template/MDD.md` — the blank template, grounded in the regulation
4. Draft `governance/how-to-fill-out.md` — annotated guidance on each section
5. Generate all five examples using Claude Code — fictional scenarios, fully filled out
6. Write `CLAUDE.md` — the engine that tells Claude Code how to use the kit
7. Write `governance/update-process.md` — regulatory monitoring and versioning process
8. Write `CHANGELOG.md` and `CONTRIBUTING.md`

9. Write `README.md` — audience, problem, quick start, submodule instructions, references
 10. Tag `v1.0.0` — first release
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How to Use This Brief in Claude Code

When starting a Claude Code session to build this repo:

```
Read project-brief.md first, then let's start building the repo structure
following the build order defined in that file.
```

Start with step 2 — pulling the SR 11-7 governance documents into the `governance/` folder.

The Federal Reserve SR 11-7 document is publicly available at:

<https://www.federalreserve.gov/supervisionreg/srletters/SR1107.htm> The 2026 revised guidance PDF is at: <https://www.federalreserve.gov/supervisionreg/srletters/SR2602.pdf>